ITMDA0-33 Project

Site: Eduvos LMS
Course: Database Management (MySQL) Assessments
Book: ITMDA0-33 Project
Printed by: Kgothatso Masha
Date: Monday, 28 September 2026, 9:29 AM

Table of contents

1. Project Specifications
2. Instructions to Students
3. Resource Requirements
4. Delivery Requirements
5. Section A
6. Terms of AI use for this Assessment
    - 6.1. Key System Features
    - 6.2. Documentation
    - 6.3. How AI use Affects Marking

1. Project Specifications

Project Specifications

Faculty:	Information Technology
Module Code:	ITMDA0-33
Module Name:	Database Management (MySQL)
Content Writer:	Clayton Nkosi
Internal Moderation:	Primrose Mthethwa

Copy Editor:	Ms Phindile Tabata
Total Marks:	100
Submission Week:	Week 6

This module is presented on NQF level 5.

5% will be deducted from the student's assignment mark for each calendar day the assignment is submitted late, up to a maximum of three calendar days. The penalty will be based on the official campus submission date.

Assignments submitted later than three calendar days after the deadline or not submitted will get 0%. [1]

This is individual project.

This project contributes 30% towards the final mark.

[1] Under no circumstances will assignments be accepted for marking after the assignments of other students have been marked and returned to the students.

2. Instructions to Students

Instructions to Students

1. Please ensure that you use the Assignment Template when preparing your submission (where applicable).
2. Please ensure that your answer file (where applicable) is named as follows before submission: **Module Code – Assessment Type – Campus Name – Student Number.**
3. Remember to keep a copy of all submitted assignments.
4. All work must be typed.
5. Please note that you will be evaluated on your writing skills in all your assignments.
6. All work must be submitted through the submission link. Negative marking will be applied if you are found guilty of plagiarism, poor writing skills, or if you have applied incorrect or insufficient referencing. (See the "instructions to students" book activity before this activity where the application of negative marking is explained.)
7. You are not allowed to offer your work for sale or to purchase the work of other students. This includes the use of professional assignment writers and websites, such as Essay Box. You are also not allowed to make use of artificial intelligence tools, such as ChatGPT, to create content and submit it as your own work. If this should happen, Eduvos reserves the right not to accept future submissions from you.
8. AI may be used only for defined supporting tasks (e.g. grammar checking, initial brainstorming). Substantive analysis and writing must be the student's own. Declaration required.

3. Resource Requirements

Resource Requirements

- The Database Management (MySQL) Module Page.
- Project Resources Folder

4. Delivery Requirements

Delivery Requirements

Delivery Requirements (evidence to be presented by students) The project submission must include:

- Program Design
- Project Source Code
- User Documentation
- Database (if any)
- Resources (if any)
- Systems Requirement (if any)
- Rough Work (if any)

5. Section A

Section A

EduLaunch Career Readiness Platform

EduLaunch is a student career-development and employability initiative that helps students move from campus to career. It provides career guidance, CV and cover-letter support, interview preparation, workplace-readiness programmes, career resources and structured support from career advisers.

EduLaunch requires a structured database to manage student accounts, career advisers, career-development programmes, programme enrolments, learning activities, progress, assessments, digital career resources and system events. The platform supports preparation and access to opportunities, but it does not guarantee employment or placement.

You are hired as the Database Designer to develop a MySQL database for the EduLaunch platform.

Application Overview

EduLaunch requires a MySQL database to support its student career-development and employability services. The database will provide a structured system for managing student accounts, career advisers, career-development programmes, programme enrolments, learning activities, assessments, results and digital career resources.

Students will register accounts that include access-control and storage information. Account access must be monitored so that expired accounts can be blocked until renewed. Career advisers will manage and facilitate career-development programmes in areas such as CV support, interview preparation, career guidance and workplace readiness. Students may enrol in several programmes, provided that their accounts are active, the programme has available capacity and they are not already enrolled.

The system will also record student progress, completion status, career-readiness points, assessment marks and career-readiness scores. Digital resources such as PDF guides, videos, presentation slides, CV templates and career-planning documents will be managed while ensuring that students do not exceed their allocated storage limits. Important errors and

system events, including failed logins, unsuccessful enrolments and storage errors, must be recorded for monitoring purposes.

The project requires students to design an Entity Relationship Diagram and implement the database using suitable tables, relationships, data types and validation constraints. Students must populate the database with realistic sample data and create views, stored procedures, triggers and indexes to support reporting, automation, validation and query optimisation.

The completed database must demonstrate the ability to identify blocked accounts, top-performing students, popular programmes and frequently downloaded resources. Students must also submit a database backup, SQL script files and supporting project evidence showing that the EduLaunch database was successfully designed, implemented and tested.

AI Use Requirements for This Assessment

This assessment is classified as AI-1, as defined in the AI Use Category Table below.

AI Use Category Table

For this specific assessment, the following applies:

Code	Category	What This Means	Evidence Required	Disclosure Required
AI-1	Limited AI Use	You may use AI only for simple support tasks (e.g. checking the grammar on your final document). The main work (virtualisation, configurations, and capturing the supporting screenshot evidence) must still be your own.	In some tasks, you may also be asked for evidence of your process (checking the grammar on your final document).	You must complete and submit an **AI Disclosure Appendix** explaining how you used AI. Access the AI Disclosure Appendix Template here.

6. Terms of AI use for this Assessment

Terms of AI use for this Assessment

1. What Counts as AI?

AI includes standalone or embedded generative systems that create or materially transform assessed work.

This includes AI features embedded in browsers, search engines, office applications, development environments, coding assistants, data platforms, security tools and engineering software.

The product name is less important than what the feature does.

See What Counts as AI

2. Permitted AI Activities

AI may suggest search terms, but it may not locate, select, summarise or evaluate sources for you.

See Permitted AI Activities

3. Assessed Concepts and General Terminology

You may not use AI to explain an assessed module concept as a substitute for engaging with the approved learning material. Each assessment will distinguish between:

Category	Meaning
Assessed concepts	Module knowledge, methods, technical processes or decisions that you must understand and demonstrate independently
General academic terminology	Terms such as peer review, Boolean operator, introduction or primary source that are not themselves being assessed
Borderline terminology	A term whose status is unclear and must be confirmed with the lecturer before using AI

4. Generally Prohibited Activities

These restrictions apply only where the activity forms part of the published assessment criteria. Assessment-specific instructions take precedence where they expressly permit or provide a tool, dataset, configuration, model, code component or other output. Please refer to the below and apply as appropriate.

See Generally Prohibited Activities

4.1. Discipline-specific Prohibited Activities

Please ensure that you consult the prohibited activity list that is specific to this module's discipline from the four discipline options below.

4.1.1. Software Engineering

4.1.2. Network Security Engineering

4.1.3. Data Science

4.1.4. Robotics

6.1. Key System Features

1. Student Accounts

Each student must register an account with:

- A unique username or student number.
- A password.
- An account status of Active or Blocked.
- An EduLaunch access expiry date.
- A maximum storage limit for career documents and resources

When EduLaunch access expires, the student's account must be blocked until the access period is renewed.

2. Career Advisers

Each career adviser must have a profile containing:

- Name.
- Email address.
- Career-support specialisation, for example CV support, interview preparation or workplace readiness.

A career adviser may facilitate multiple career-development programmes.

3. Career Development Programmes

Each career-development programme must store:

- Programme title.
- Category, for example CV Support, Interview Preparation, Career Guidance or Workplace Readiness.
- Level: Foundation, Intermediate or Advanced.
- Capacity limit, representing the maximum number of students.
- Assigned career adviser.

Students may enrol in multiple career-development programmes.

4. Programme Enrolments

The database must:

- Track student enrolments for each career-development programme.
- Prevent enrolment when the programme capacity has been reached.
- Prevent duplicate enrolments for the same student and programme.

5. Learning Activities and Progress

Each career-development programme contains multiple learning activities. For each student registered for an activity, track:

- Completion status: Not Started, In Progress or Completed.
- Progress percentage from 0 to 100%.
- Career-readiness points earned.

6. Assessments and Results

Each career-development programme may contain assessments such as quizzes, CV reviews, mock interviews or practical workplace-readiness tasks. For each student, record:

- Marks obtained from 0 to 100.
- Career-readiness score gained.

7. Digital Career Resources

Resources may include PDF guides, videos, slides, CV templates, cover-letter templates and career-planning documents. Track:

- Resource type.
- Resource size in megabytes.
- Number of downloads.

Students have limited personal storage for career documents and may not add files beyond their storage limit.

8. Error and Event Logging

Log important system errors or events, including:

- Login failures.
- Programme-enrolment errors.
- Assessment-submission or resource-storage errors.

Each log record must store the student ID where applicable, error category, error message and timestamp.

System Specifications

You are required to complete the following tasks:

1. ER Diagram

Create an Entity Relationship Diagram for the EduLaunch scenario. Show all required entities, primary keys, foreign keys, relationships and cardinalities.

2. Database and Table Creation

Create a MySQL database for the EduLaunch scenario.

Create the tables required by the scenario.

Ensure that suitable data types, primary keys, foreign keys, UNIQUE, NOT NULL, DEFAULT, CHECK and other validation constraints are used to prevent invalid data and avoid losing marks.

The tables must prevent invalid percentages and marks, negative storage or capacity values, duplicate programme enrolments and invalid status values.

3. Insert Sample Data

Insert realistic sample data into the tables. Include at least five students, three career advisers, five career-development programmes, programme enrolments, learning activities, progress records, assessments, results and digital career resources.

4. Create the Following Views

vwBlockedStudents: students whose EduLaunch access has expired and whose accounts are blocked.

vwTopCareerReadyStudents: the top 20 students according to total career-readiness score.

vwMostDownloadedResources: the top 20 career resources according to download count.

vwPopularCareerProgrammes: the top five career-development programmes according to enrolment count.

5. Create the Following Stored Procedures

spRegisterStudent - registers a new student and ensures that the student number or username is unique.

spRenewAccess - adds days to a student's access period and reactivates a blocked account.

spEnrolStudent - enrols a student only when the programme has available capacity, the account is active and the student is not already enrolled.

spAddCareerResource - adds a resource to a student's personal library only when the storage limit will not be exceeded.

spSendAccessNotice - produces notices showing the remaining EduLaunch access time for students.

6. Create Triggers

Automatically block student accounts when EduLaunch access expires.

Log errors or events when important system actions occur, such as failed enrolments, failed resource additions or assessment-submission errors.

7. Create Indexes

Create suitable indexes for faster lookups on students, career-development programmes and programme enrolments.

8. Database Backup and Submission

Create a zipped backup or SQL dump of the completed EduLaunch database and submit it together with the SQL script files and project evidence.

6.2. Documentation

You must submit proper user and project documentation, which includes:

- An explanation of how the system works
- Screenshots of code and system output
- A brief user guide explaining how to use the system
- A description of the database structure, including tables and fields
- A description of the classes, sessions, validation and file-handling features used

6.3. How AI use Affects Marking

How AI use Affects Marking

Finding	Outcome
No AI used and accurately declared	Your lecturer will mark according to the marking guide.
Permitted AI use accurately declared	Your lecturer will mark according to the marking guide.
AI-generated in identifiable assessed component	Your lecturer will mark authentic work and will not award marks for the affected component that is identified to be AI-generated.
AI-generated in dependent criteria may be affected	Your lecturer will evaluate each criterion in isolation and will not award marks for affected criteria that are AI-generated.
Technically correct work that raises authenticity concerns*	Your lecturer will complete the review before finalising the mark, and will not award marks for affected criteria that are AI-generated.
Serious authenticity concerns exist*	Your lecturer may request a formal discussion with you, and depending on the outcome may escalate a case for formal academic-integrity hearing.

*See General Prohibited Activity list under Terms of AI use for this Assessment